

Yousuf Abdelhady

yusufshehabi0611@gmail.com | +201033267098 | Cairo, Egypt

in LinkedIn | GitHub | codeforces

EDUCATION

Ain Shams University, Faculty of Computer and Information Sciences
Bachelor's in Data Science

Cairo, EG
Oct 2027

- Relevant Coursework: Object-Oriented Programming, Data Structures, Algorithms Analysis & Design, Databases, Artificial Intelligence, Computer Networks, Compiler Design, Software Analysis & Design.
- GPA: **3.65/4.00**

VOLUNTEER EXPERIENCE

acmASCIS

Cairo, Egypt
Sep 2025 – Present

Training Head / Competitive Programming Instructor

- Led the competitive programming training program for college students across multiple skill levels.
- Mentoring students of all levels in my college through both offline and online sessions.
- Designed and organized structured training plans covering topics from introductory programming to advanced areas, including Number Theory, Graphs, Dynamic Programming, and Data Structures.
- Supervised and mentored instructors and problem setters within the training team.
- Guiding students in problem-solving techniques and preparing them for competitive programming contests.
- Prepared students for participation in local and regional programming contests.

SKILLS

- **Programming Languages:** C++, C#, Java, Python, Dart
- **Competitive programming:** 2000+ solved problems and Expert rating on Codeforces.
- **Soft skills:** Mentoring, Managing, problem-solving, and teaching.

HONORS AND AWARDS

ECPC

Silver Medalist

Aug - 2025

- Ranked as one of the top **10** teams in Egypt over **+250** participating teams and qualified as one of the Egyptian teams representing Egypt in the **2025 ICPC Egyptian Collegiate Programming Contest (ECPC)**.

ACPC

Finalist

Aug - 2025

- Competed among **150** regional teams in the **2025 ICPC Africa & Arab Collegiate Programming Championship (ACPC)**, representing Egypt at the regional level.

PROJECTS

Image Segmentation

May - 2025

- Implemented **graph-based segmentation** using **disjoint set union (union-find)** and **Kruskal's MST** for region merging with 8-connected neighborhoods.
- Optimized edge processing with a **counting sort algorithm** for **O(n)** edge weight sorting and **parallelized** R/G/B channel segmentation.
- Built a **multi-channel processing** system with **parallel channel segmentation** and intersection logic for color image handling.

Mini-Wasalny

May - 2025

- Built a Qt desktop application for **visualizing graph algorithms (DFS, BFS, Dijkstra, Floyd-Warshall)**.
- Implemented **interactive graph manipulation** with node dragging, edge weight editing, and step-by-step algorithm execution. with real-time color highlighting.
- Developed a **user authentication system** with **JSON-based persistent storage** for credentials and graph state.

Watch-It

Nov - 2024

- Developed a **full-stack Java application** simulating streaming platform functionality with user/admin roles and interactive UI.
- Implemented a search system using **dynamic programming** for efficient movie/TV show lookup.
- Designed a responsive **JavaFX interface** with modern UI components and detailed media information displays.
- Built a user management system with **regex validation** for credentials and personalized watchlist/rating features.